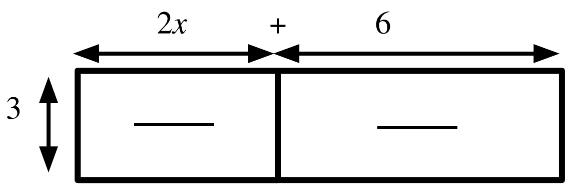
Name: ______________________ **Worksheet**

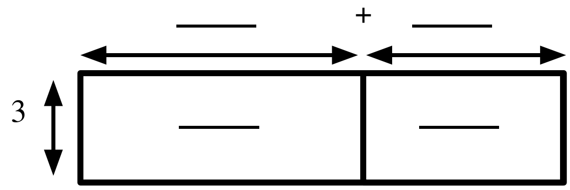
Checking and securing understanding of the distributive law with algebraic terms

Task A: Multiplying expressions by a term

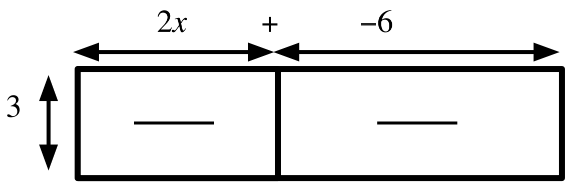
1) Use the area models to write these expressions in expanded form



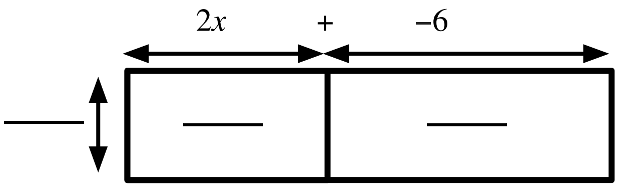
a) $3(2x + 6) \equiv$ _____



c) $3(6 - 2x) \equiv$ _____

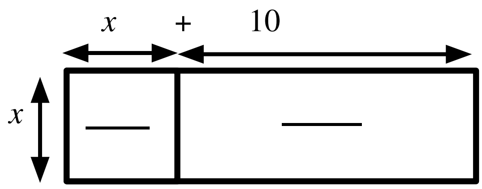


b) $3(2x - 6) \equiv$ _____

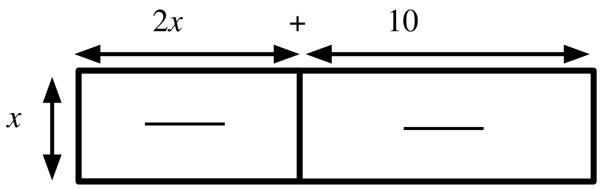


d) $-3(2x - 6) \equiv$ _____

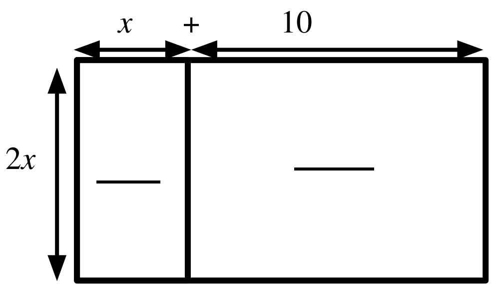
2) Use the area models to write these expressions in expanded form



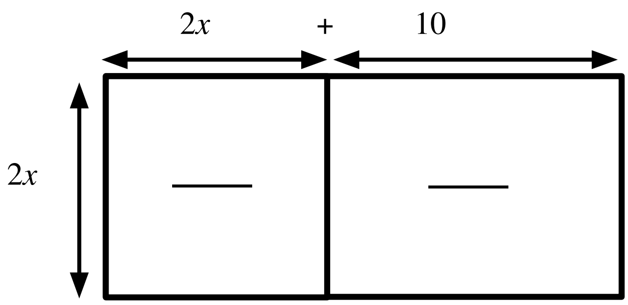
a) $x(x + 10) \equiv$ _____



c) $x(2x + 10) \equiv$ _____



b) $2x(x + 10) \equiv$ _____



d) $2x(2x + 10) \equiv$ _____

3) Write each of these in expanded form.

a) $5(x + 4)$

b) $5(2x + 4)$

c) $5(2x - 4)$

d) $10(2x - 4)$

e) $10(4 - 2x)$

f) $10(-2x + 4)$

g) $-10(-2x + 4)$

h) $5x(x + 4)$

i) $5x(2x + 4)$

j) $5x(2 + 4x)$

k) $5x(2 - 4x)$

l) $-5x(2 - 4x)$

m) $-5x(2 - 4x + y)$

n) $-5x(2 - 4x) + y$

Task B: Multiplying and simplifying multiple expressions

1) Expand and simplify where possible.

a) $4(2x + 6) + 5(x + 2)$

b) $7(x - 5) + 2(x + 13)$

c) $x(2x + 1) + 6(5x - 3)$

d) $5x(3x + 7) - x(3x + 7)$

e) $4x(2y + 7) - 2(3x - 4)$

f) $3(4y + 7x) - 2(4y + 7x)$

g) $-5x(8y - 7x) - (yx - 10x)$

2) All four of these expressions are equivalent to each other.
Fill in the missing terms.

a) $7(x - 3) + 4(2x + 6.5)$

b) $2(3x + 4) + 3(3x + \boxed{\quad})$

c) $5(\boxed{\quad} - 1) - 5(x - 2)$

d) $7(3x + 1) - \boxed{\quad}(3x + 1)$